1. Introduction

This project proposes a jurisdiction-aware RegTech tool designed to monitor fairness and compliance risks in AI-driven credit scoring systems.
The tool is designed for multinational financial institutions operating across jurisdictions with differing regulatory expectations regarding AI governance, explainability, and fair lending practices.
The project focuses on the differences between the United States and the European Union regulatory environments, particularly in the context of AI-based credit decision systems.

2. Selected Regulated Entity: HSBC Holdings

HSBC Holdings is a multinational banking and financial services organization operating across major global jurisdictions including the United States, the United Kingdom, the European Union, Hong Kong, and Singapore.
HSBC was selected because it operates across highly diverse regulatory environments and increasingly relies on digital systems and AI-assisted processes in areas such as customer onboarding, fraud detection, and credit risk assessment.
As a globally regulated financial institution, HSBC faces significant challenges in maintaining regulatory consistency while adapting to different local compliance requirements.

3. Selected Regulatory Domain: AI Credit Scoring and Fair Lending Governance

The selected regulatory domain for this project is AI-driven credit scoring and fair lending governance.
Financial institutions increasingly use machine learning models to automate or assist lending decisions. While these systems may improve efficiency and predictive performance, they also introduce risks related to algorithmic bias, discrimination, explainability, and model governance.
Regulators across jurisdictions have adopted different approaches to managing these risks. As a result, multinational financial institutions must ensure that AI systems satisfy multiple and sometimes conflicting regulatory expectations.

4. Why This Problem Matters

AI-driven lending systems can unintentionally produce discriminatory outcomes against demographic groups based on gender, ethnicity, age, or socioeconomic status.

These risks are especially significant in consumer lending because lending decisions directly affect financial access and economic opportunity.

Traditional compliance processes based on spreadsheets or static reporting are often insufficient for monitoring continuously evolving machine learning systems. This creates demand for RegTech solutions capable of real-time fairness monitoring, explainability analysis, and jurisdiction-specific governance controls.

5. Regulatory Divergence Between the United States and the European Union

The United States and the European Union adopt different regulatory philosophies regarding AI governance and fair lending.

In the United States, regulators such as the Consumer Financial Protection Bureau (CFPB) emphasize explainability and adverse action transparency under frameworks such as the Equal Credit Opportunity Act (ECOA) and Regulation B. Financial institutions must be able to explain why a consumer was denied credit and demonstrate that discriminatory lending practices are not occurring.

In contrast, the European Union adopts a broader risk-governance approach through the EU AI Act. AI-based credit scoring systems are categorized as high-risk AI systems and are subject to extensive governance requirements, including bias testing, risk management, data quality controls, documentation, and human oversight obligations.

As a result, multinational financial institutions must maintain compliance with multiple regulatory expectations simultaneously.

Area	United States	European Union
Main Focus	Explainability	Risk Governance
Key Regulation	ECOA / Regulation B	EU AI Act
Compliance Concern	Adverse Action Notice	High-Risk AI Oversight
Fairness Requirement	Anti-discrimination	Bias Testing
Governance Style	Consumer protection	Systemic AI governance

6. Business Motivation

A jurisdiction-aware RegTech platform can help multinational banks reduce regulatory risk, improve model governance, and strengthen operational transparency.
The proposed tool enables institutions to monitor fairness metrics, apply jurisdiction-specific thresholds, generate explainability outputs, and identify compliance risks before regulatory violations occur.
This creates both operational and strategic value by reducing compliance costs, improving governance efficiency, and strengthening institutional trust.
The proposed RegTech tool follows a multi-stage compliance monitoring workflow integrating machine learning, fairness evaluation, and jurisdiction-specific governance logic.

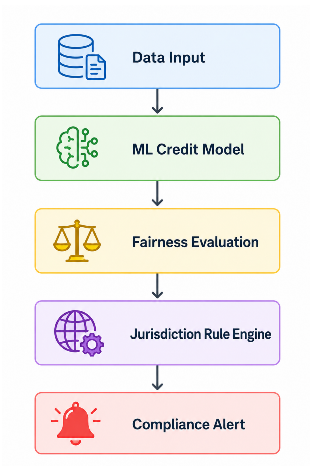

Figure 1. High-level architecture preview of the proposed jurisdiction-aware RegTech tool.

7. Conclusion

The increasing use of AI-driven credit scoring systems creates significant regulatory and governance challenges for multinational financial institutions.

Differences between US and EU regulatory approaches require organizations to adopt flexible and jurisdiction-aware compliance solutions capable of adapting to varying fairness, explainability, and governance expectations.

This project therefore proposes a RegTech solution that combines fairness monitoring, explainability analysis, and jurisdiction-specific governance controls to support more responsible and compliant AI-enabled lending practices.

8. References

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